Lasers in Medical Science (2025) 40:31
https://doi.org/10.1007/s10103-024-04242-5

ORIGINAL ARTICLE

The efficiency of single-session photobiomodulation on healing of periapical bone lesions using CBCT: a randomized clinical trial

Hend Hamdy Ismail[1] · **Maram Farouk Obeid**[1] · **Ehab ElSaid Hassanein**[1]

Received: 16 April 2024 / Accepted: 13 November 2024 / Published online: 23 January 2025
© The Author(s) 2024

Abstract

As photobiomodulation is growing in the dental field the aim of this prospective, two-arm clinical trial was to assess the radiographic changes for chronic periapical bone lesions related to mandibular molars after primary root canal therapy with or without applying Diode laser on soft tissue. The samples were randomly divided into a Laser group and a mock laser (ML) group. Preoperative CBCT images were compared 12 months later with postoperative CBCT to gauge the changes in the volume of the bony lesion by two observers. The kappa coefficient was calculated to assess the intra-observer agreement. Data were presented as median, mean, and standard deviation (SD) values. For parametric data (age), a Student's t-test was used. Qualitative data (Gender) was presented as frequencies and percentages and then analyzed by Chi-square test. The Wilcoxon signed-rank test was used to compare CBCT measurements pre- and post-treatment. The significance level was set at $P \leq 0.05$. The collected data in this study showed that there was no statistically significant difference between median volumes of the periapical lesion in the Laser and ML groups (P-value = 0.564, Effect size = 0.237) and (P-value = 0.452, Effect size = 0.310), respectively while, there was a statistically significant decrease in the volume of periapical lesion post-treatment within each group (P-value < 0.001). The size of the periapical bony lesion decreased significantly after conventional endodontic treatment whether Laser was applied or not. The study protocol was enrolled as NCT04311905 at www.clinicaltrials.gov database on (17/03/2020) after the approval of the Ethics Committee, Faculty of Dentistry, Ain Shams University (FDASURecID091703).

Keywords Soft tissue laser · Photobiomodulation · PBM · Bone healing · Periapical lesion · CBCT · Randomized clinical trial

Abbreviations

PBM	Photobiomodulation
CBCT	Cone beam Computed tomography
ML	Mock Laser
ATP	Adenosine Triphosphate

Introduction

Chronic periapical bony lesions are primarily caused by microorganisms of endodontic origin and their byproducts [1]. This condition is frequently the result of an inflammatory immune reaction, which can be identified by periapical radiography [2]. Effective treatment of root canal infections is essential for repairing periapical lesions and reducing associated inflammation, typically addressed through conservative orthograde endodontic treatment [3]. Success in this treatment relies heavily on proper disinfection of the root canal spaces through mechanical preparation and the use of appropriate irrigating solutions to reduce bacterial presence [4]. This conservative therapy can result in complete to partial apical healing of lesions, with a success rate of up to 86% [5]. Consequently, there is intense debate on how to enhance healing kinetics to improve recovery and prognosis [6].

Since its development by Maiman in 1960, laser technology has been frequently used in dentistry [7]. An area of growing interest is photobiomodulation (PBM) or low-level laser therapy (LLLT), which has gained attention due to its ability to penetrate tissues, promote healing, and reduce

✉ Maram Farouk Obeid
maramobeid@dent.asu.edu.eg

[1] Faculty of Dentistry, Ain Shams University, Organization of African Unity Street, Cairo, Egypt11766

pain without significant side effects [8]. PBM uses a red or near-infrared laser with an energy output of up to 500 mW and a wavelength range of 600–1000 nm [9–11]. In endodontics, the key element of PBM is its ability to stimulate tissue repair through cellular biostimulation [12]. This process does not heat soft tissue; instead, the energy from absorbed photons generates photochemical, photophysical, and photobiological effects [13]. When an appropriate dose of laser light interacts with cells, functions such as mitochondrial ATP synthesis can be enhanced, accelerating proliferative processes [14]. Moreover, this coherent light beam can increase local circulation and collagen synthesis [15]. Previous ex-vivo investigations have demonstrated that low-power lasers significantly impact osteoblast differentiation, proliferation, and calcification [16]. Additionally, it has been shown to promote bone formation after tooth removal by creating local conditions that accelerate bone growth and edema resolution [17].

Few in-vivo investigations have highlighted the benefits of PBM in the healing of periapical bone lesions in molars. This study aims to assess the outcomes for a group of patients with chronic periapical bone lesions related to mandibular molars who underwent primary root canal therapy with or without Diode Laser application. This prospective, two-arm, non-significant-risk clinical study evaluated the healing of periapical lesions by comparing the volume of the bony lesion on CBCT images taken before and 12 months after conventional root canal treatment. The null hypothesis posits that there is no discernible difference in the healing process when Laser is applied.

Methods

Study aim, design, and setting

The aim was to assess the outcomes for a group of patients with chronic periapical bone lesions; related to mandibular molars; who underwent primary root canal therapy with or without diode Laser by measuring the size of the bone defect before and after interventions on CBCT.

The design of this study adheres to the Consolidated Standards of Reporting Trials declaration CONSORT [18]. It is a prospective, two-arm, non-significant risk clinical, single-blinded, randomized, placebo-controlled clinical trial. The subjects were classified as presented in Fig. 1.

Ethics and consent to participate

With the identifier NCT04311879 (17/03/2020), the research protocol was enrolled into the clinical trials database at www.clinicaltrials.gov. The Ethics Committee of the Faculty of Dentistry of Ain Shams University reviewed it and gave its approval (Approval code: FDASURecID091703). All methods were performed per the relevant guidelines and

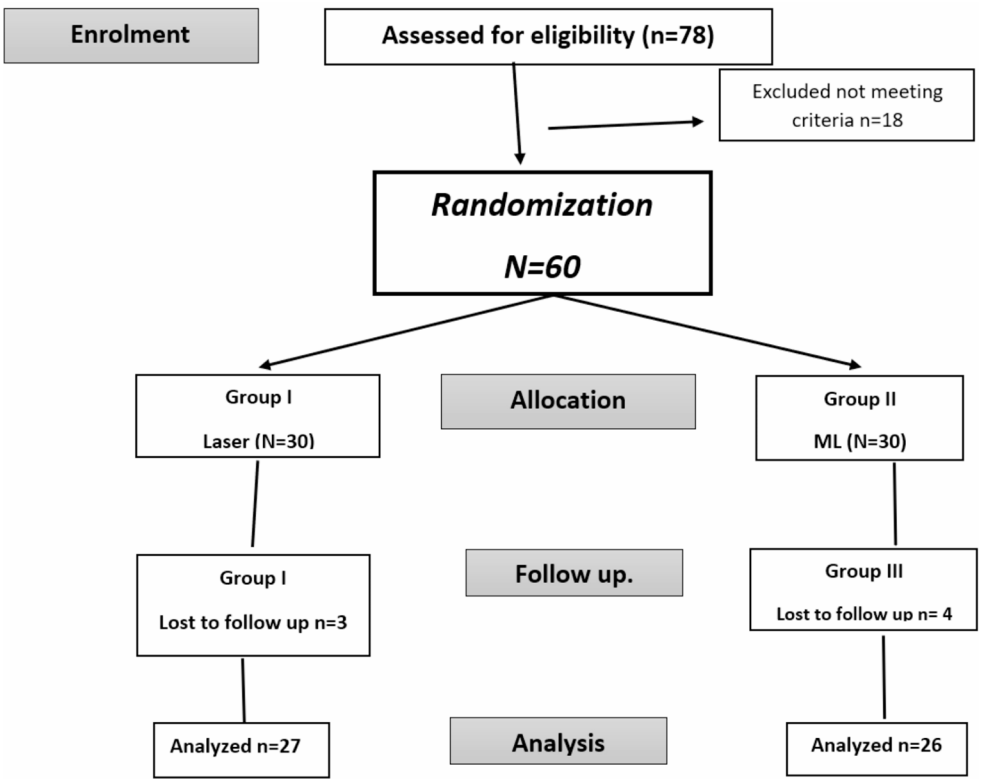


Fig. 1 Study CONSORT flow diagram

regulations of the Consolidated Standards of Reporting Trials declaration CONSORT [18].

Participants for the study were selected from the outpatient endodontic clinic at the Faculty of Dentistry at Ain Shams University, between July 2017 and January 2020. Each applicant was given a detailed explanation of the study's objectives, methodology, benefits, and potential risks. All participants who were included in the study signed a consent form.

Sample size calculation

This was estimated with referral to calculations and guidance of data from an earlier study [19]. It was estimated using Power 3 and it was built on the healing assessment of periapical lesions utilizing CBCT after one year. The alpha error was set at 0.05 and the beta error at 0.1 (power 90%). The sample size was 27 patients per group and shifted to 30 to compensate for the dropouts. The sample size was calculated on IBM SPSS Statistics (Windows, Version 23.0. Armonk, NY: IBM Corp).

The characteristics of participants (eligibility criteria)

The participants in our study were healthy individuals between the ages of 25 and 40 who had one radiographic apical radiolucency (less than 1500 mm^3) related to mandibular permanent teeth. Periapical radiographs were done to exclude teeth with immature apex, root fracture, prior root canal, or required considerable prosthetic rehabilitation. Additionally, patients with widespread periodontal disease, pregnant or nursing women, and those with a pre-existing medical condition or dental issue that put them at risk during the experiment or those with any systemic diseases that may affect the wound-healing process, as well as heavy tobacco users, were also banned from participation.

Diagnosis and patient acceptance

An experienced endodontist (primary researcher) carried out all aspects of dental management. After detailed medical and dental history, clinical and radiographic assessment was done to establish an accurate diagnosis. The presence of periapical lesions was documented by having a preoperative CBCT imaging (90 kVp, 12 mA) applying 0.75 mm^3 voxel size. The time of exposure was 26 s.

Out of the 78 patients initially considered for the study, only 60 were included. The remaining 18 patients were excluded due to various reasons: 6 patients had underlying systemic health problems, such as uncontrolled diabetes, which could affect the healing process; 3 patients were heavy smokers; 2 patients had previously undergone root canal treatment on the targeted teeth; 2 patients refused to participate; and 1 patient was excluded due to pregnancy.

Root canal treatment

Under rubber dam isolation, endodontic access was made after the administration of local anesthesia (1:80,000 Arcaine, Aarge Pvt Ltd., India). The working length estimated by an apex locator (Root ZX mini, J. Morita, Japan) was verified by a periapical radiograph. The rotary nickel-titanium Protaper Gold system (Dentsply Maillefer, Ballaigues, Switzerland) was used to mechanically prepare canals by the manufacturer's instructions. Depending on the canal's original diameter, finishing files ranging from F1 to F5 mounted on a torque-controlled endodontic motor (X-Smart; Dentsply Sirona) were used. A 27-gauge needle attached to a 3 mL syringe (Kendall, Covidien, Mansfield, MA, USA) was used to irrigate between the files. The irrigation solution used was 3 mL of 2.5% NaOCl. This was followed by a final rinse of 3 mL of 2.5% NaOCl, 3 mL of EDTA 17% (Denteck, Zoetermeer, Netherlands), and 3 mL of ordinary saline in between and at the end before filling the canal.

All canals were dried out with paper points and packed with gutta-percha and AH Plus sealer (Dentsply Maillefer, Ballaigues, Switzerland) applying the warm vertical condensation technique. All teeth were restored with composite resin (Dentsply Sirona, Charlotte, NC, USA) at the same visit. A dental operating microscope (Carl Zeiss, Jena, Germany) X8 magnification was used throughout the treatment.

Randomization

Following that, patients were divided into two groups ($n = 30$) based on how the diode laser was used. The researcher made envelopes with secret group codes progressively distributed to the qualified patients. To have sufficient randomization utilizing an equal proportion randomization allocation ratio, the operator opened the envelopes and logged the codes with the appropriate patient characteristics on a computer. The patient was blinded to the group he or she enrolled in.

Intervention

- *In the Laser group*, the primary researcher positioned the laser applicator over the oral mucosa with the beam concentrated apically. The PBM Parameter is presented in Table 1 [20–22].

Table 1 PBM parameters

Parameter	Laser Group	ML Group
Manufacturer	LITEMEDICS	
Model identifier	Serial Number: 148, SwvM. 150VS108VT.100	
wavelengths	970 nm	
Pulse mode	Continuous, 10 Hz	
Beam Spot size at target	Around 0.3 mm	
Exposure Duration	60 s per point	
Number of points irradiated	Two points at the level of the targeted root apex buccally and lingually	
Area irradiated	Around 0.7 mm2 (for both points)	0 (device turned off)
Application technique	Without movement in a contact mode	
Power:	0.5 W	0 W (device turned off)
Energy per Point:	30 J	0 J (since the laser is off)
Total irradiation Energy:	60 J	0 J (since the laser is off)
Power Density:	2.86 W/cm²	0 W/cm²
Total energy	60 J	0 J
Number & frequency of treatment sessions	Single session immediately after RCT	

- *In the ML group*, the diode laser tip was positioned similarly as in the Laser group but without laser activation by the same researcher.

Recall rate of patients

Each participant was recalled after one year, the recall rate for this study was 88.3%. Out of 60 patients who got treatment, 4 patients in the ML group and 3 patients in the Laser group with a total of 7 patients did not return owing to personal reasons, thus, 53 patients returned for follow-up care and undertook post-operative CBCT using the same machine and the same parameters used in the preoperative scan.

Assessment of periapical Healing

Two different, skilled examiners, who were concealed from the groups, performed the measures to guarantee their reliability using Planmeca software (Planmeca Romexis®; Planmeca). All CBCT pictures were analyzed on a 27-inch flat-panel computer display with a pixel resolution of 2.560 × 1.440. Brightness and contrast were the only adjustments made after the filters were set to normal.

For both the preoperative and postoperative CBCT scans, the volume of the periapical lesion was measured in mm^3. The investigator manually traced the shape of the lesion in the axial cut using the Romexis software's manual segmentation application, repeating the process every 2 mm, until the lesion vanished from the bone and healthy bone was visible. The lesion volumetric region was then determined by selecting "create region" in the application. The software calculated all the data.

Statistical analysis

The kappa coefficient was calculated to assess the agreements between the observers. The numerical data was explored for normality by checking the distribution of data and using tests of normality (Kolmogorov-Smirnov and Shapiro-Wilk tests). Age data showed a normal (parametric) distribution while CBCT data showed non-normal (non-parametric) distribution. Data were presented as median, mean, and standard deviation (SD) values. For parametric data, Student's t-test was used for comparisons between age values in the two groups. For non-parametric data, the Mann-Whitney U test was used to compare between the two groups. The Wilcoxon signed-rank test was used to compare CBCT measurements pre- and post-treatment. Qualitative data (Gender) was presented as frequencies and percentages. The chi-square test was used for comparisons between the groups. The significance level was set at $P \leq 0.05$. Statistical analysis was performed with IBM SPSS Statistics for Windows, Version 23.0. Armonk, NY: IBM Corp.

Results

Out of 78 patients, 60 were deemed qualified after being evaluated according to the predefined inclusion criteria. All participants were recruited after one year, 3 patients from the Laser group and 4 in the ML group failed to follow up, resulting in a final radiographic assessment of 53 patients (27 in the Laser group and 26 in the ML group). None of the participants showed any signs/symptoms of failure clinically.

The agreement for volumetric measurements between observers was found to be excellent (kappa > 90).

Regarding the demographic data (summarized in Table 2), there was no statistically significant difference between mean age values in the two groups. As regards gender; the Laser group showed a statistically significantly higher prevalence of males and a lower prevalence of females than the ML group.

Regarding the comparison between the two groups (presented in Table 3; Figs. 2 and 3); there was no statistically significant difference between median volumes of the periapical lesion in the two groups before treatment (P-value = 0.564, Effect size = 0.237). Also, after the

Table 2 Mean, standard deviation (SD), frequencies (n), and percentages (%) of demographic data in the two groups

Demographic data	Laser Group ($n=27$)	ML Group ($n=26$)	*P*-value
Gender [n (%)]			**0.001***
Male	**21 (77.8%)**	**14 (53.9%)**	
Female	**6 (22.2%)**	**12 (46.1%)**	
Age (Years)			
Mean (SD)	**38 (9.6)**	**37.6 (9.5)**	**0.826**

*: Significant at $P \leq 0.05$

Table 3 Comparison between the volume of the periapical lesion (mm^3) in the two groups

Time	Laser Group	ML Group	*P*-value	Effect size (d)
Pre-treatment				
Median	**50.5**	**77**	**0.564**	**0.237**
Mean (SD)	**160.1 (311.8)**	**91.4 (72.8)**		
Post-treatment				
Median	**19**	**10.5 (2–83)**	**0.452**	**0.310**
Mean (SD)	**78.4 (155)**	**20 (23.3)**		
P-value	**0.002***	**0.002***		
Effect size (r)	**0.883**	**0.884**		

*: Significant at $P \leq 0.05$

intervention, there was no statistically significant difference in the groups (P-value = 0.452, Effect size = 0.310).

Regarding the changes within each group (presented in Table 3); there was a statistically significant decrease in the volume of periapical lesions post-treatment in each group (P-value = 0.002, Effect size = 0.883) and (P-value = 0.002, Effect size = 0.884), respectively.

Discussion

PBM is a type of phototherapy in which a monochromatic, coherent light of low intensity is used. In several fields of medicine and dentistry, it has emerged as a viable therapy option. Even though there are numerous studies demonstrating the benefits of PBM [7–9], its impact on periapical lesion healing is not clear. Thus, we aimed in our clinical study to test the effects of using Laser following conventional endodontic therapy on bone healing in terms of volume measured by CBCT in mandibular molars with periapical lesions.

This study is a randomized clinical trial (RCT); the most effective resource in evidence-based medicine. It was chosen as it accurately reflects the clinical aspect with the highest dependability [23].

It is known that the primary goal of photobiomodulation (PBM) is to achieve therapeutic effects without inducing thermal damage to tissues (non-thermal biostimulation) [10]. In our investigation, we chose a Diode laser with 970 nm wavelength. This wavelength is efficiently absorbed by water and hemoglobin, which facilitates increased local circulation, enhanced oxygen delivery, and improved cellular metabolism [24]. While other wavelengths (e.g., 660 nm, 810 nm, 1064 nm) have also been shown to produce biostimulatory effects, the choice of wavelength often depends on the desired depth of tissue penetration. For instance, the 660 nm wavelength is effective for superficial tissue applications, whereas wavelengths in the near-infrared range, such as 810 nm and 980 nm, are better suited for deeper tissue penetration, which is necessary for the studied effect, as the laser aims to impact bone repair in the periapical area [25]. The laser beam was targeted over the alveolar mucosa at a location corresponding to the mesial and distal apices of each tooth both buccally and lingually to be aligned with previous studies [20–22, 26]. Given that it enables blinding in clinical research, a mock laser was used to serve as a placebo in this case [27]. The healing of periapical bone lesions was measured in terms of mm^3 on CBCT as it is proved to be more accurate at detecting this than digital periapical radiographs in the long-term assessment of root canal treatment success [28]. Furthermore, CBCT was quite accurate in determining the volume of periapical pathosis [29].

Our null hypothesis was accepted, and the results showed that there is no discernible difference in the healing process when Laser is applied. The size of the bony lesions decreased after conventional root canal treatment, and this was in line with a previous study [30]. Our demographic data revealed the significant presence of males in the Laser group. However, it was previously concluded that gender had no significant effect on bone healing [31–33].

While we recognize the importance of comparing our findings with previous studies, it is important to note that there is a lack of articles specifically investigating bone healing after endodontic treatment with photobiomodulation (PBM). As a result, direct comparisons with other studies are not feasible. However, formerly, it was demonstrated that PBM is effective in wound healing [34] as well as bone healing [35]. It could improve callus formation in the early phase of the bone healing procedure and may speed up bone growth and edema resolution [17, 36]. This positive effect of PBM marked by decreasing the volume of the bony lesion can be credited to the impact of the laser beam [8, 37, 38]. Usually, this leads to non-thermal effects with an increase in the generation of ATP, the "energetic currency" for cells, and accelerates cells' different functions [14]. Additionally, PBM is recognized as a non-invasive kind of light therapy (phototherapy) [16]. It affects cellular metabolism as well as local microcirculation [15]. It increases prostaglandin synthesis providing potent anti-inflammatory effects [39]. Likewise, PBM promotes oxidation/reduction in cells by activating the mitochondria, leading to enhanced ATP generation and protein synthesis, which can hasten periradicular

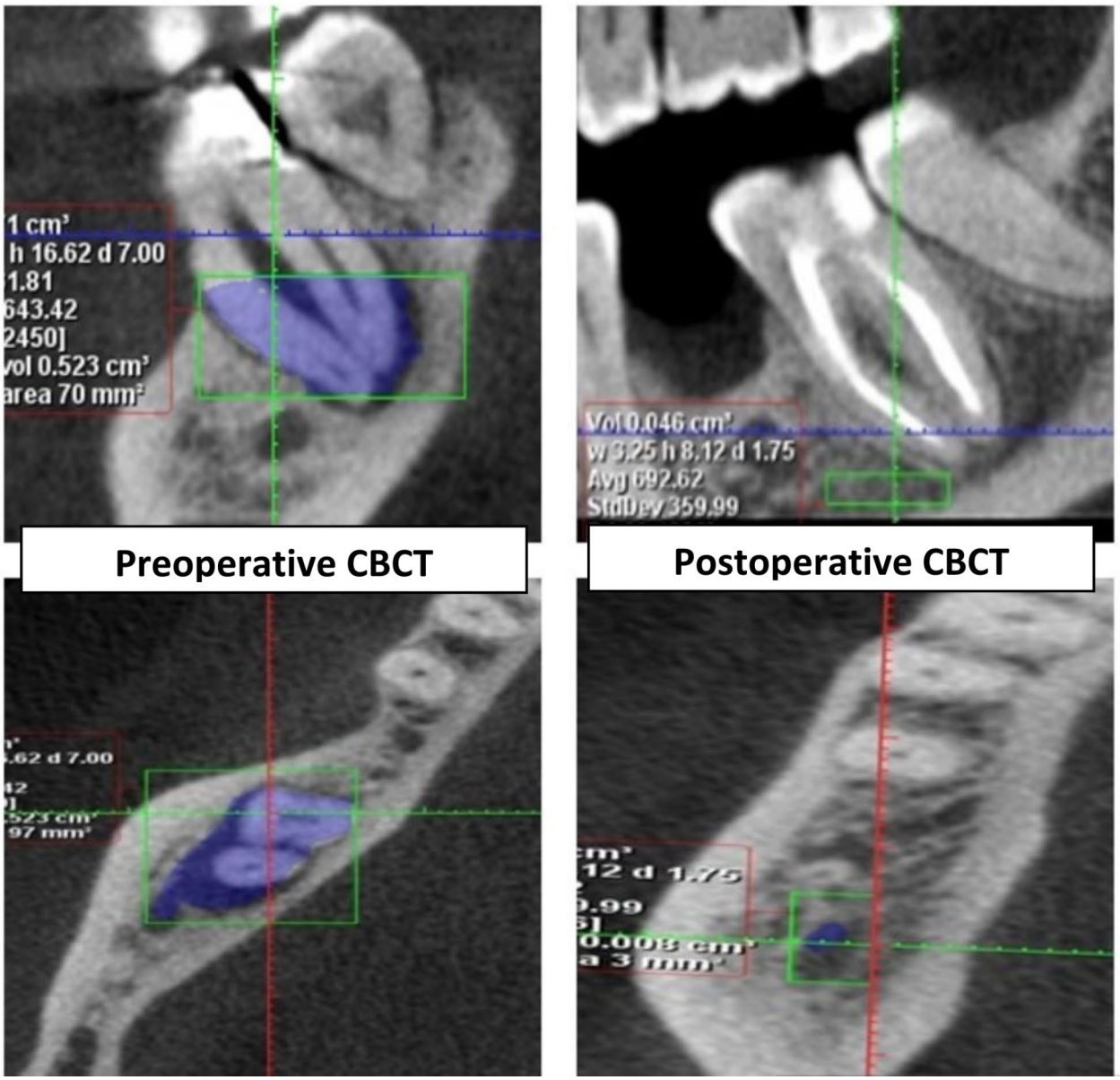


Fig. 2 CBCT images showing the difference in the volume of the periapical lesion in the Laser group

healing after root canal treatment [40]. All these biological processes may account for the promising effect on healing as reported earlier [34–36, 41].

The statistical lack of significance between the Laser and ML groups in our outcomes might be attributed to several factors. Firstly, the timing of the post-treatment evaluation may have impacted the results; our 12-month follow-up period may not have been sufficient to capture the full extent of the healing process facilitated by PBM. Longer follow-up periods are often necessary to observe significant differences in tissue healing and regeneration [25]. Moreover, the mode of laser application in this trial could also play a role. The laser was used only once immediately following endodontic therapy. In the literature, there is a wide range of PBM protocols for bone healing; from daily to alternate-day administration for 15–21 days [42]. This study used a single session as per the protocols followed in previous investigations [21, 43, 44]. However, this can be considered a limitation of this study. As a previously published systematic review [9] suggests, more research needs to be conducted with varied application protocols to validate the effectiveness and potential risks associated with PBM in endodontics. Nonetheless, it can be an effective complementary tool to traditional treatment.

After all, there is a terrific demand to establish an efficient protocol of laser use concerning wavelength, power output,

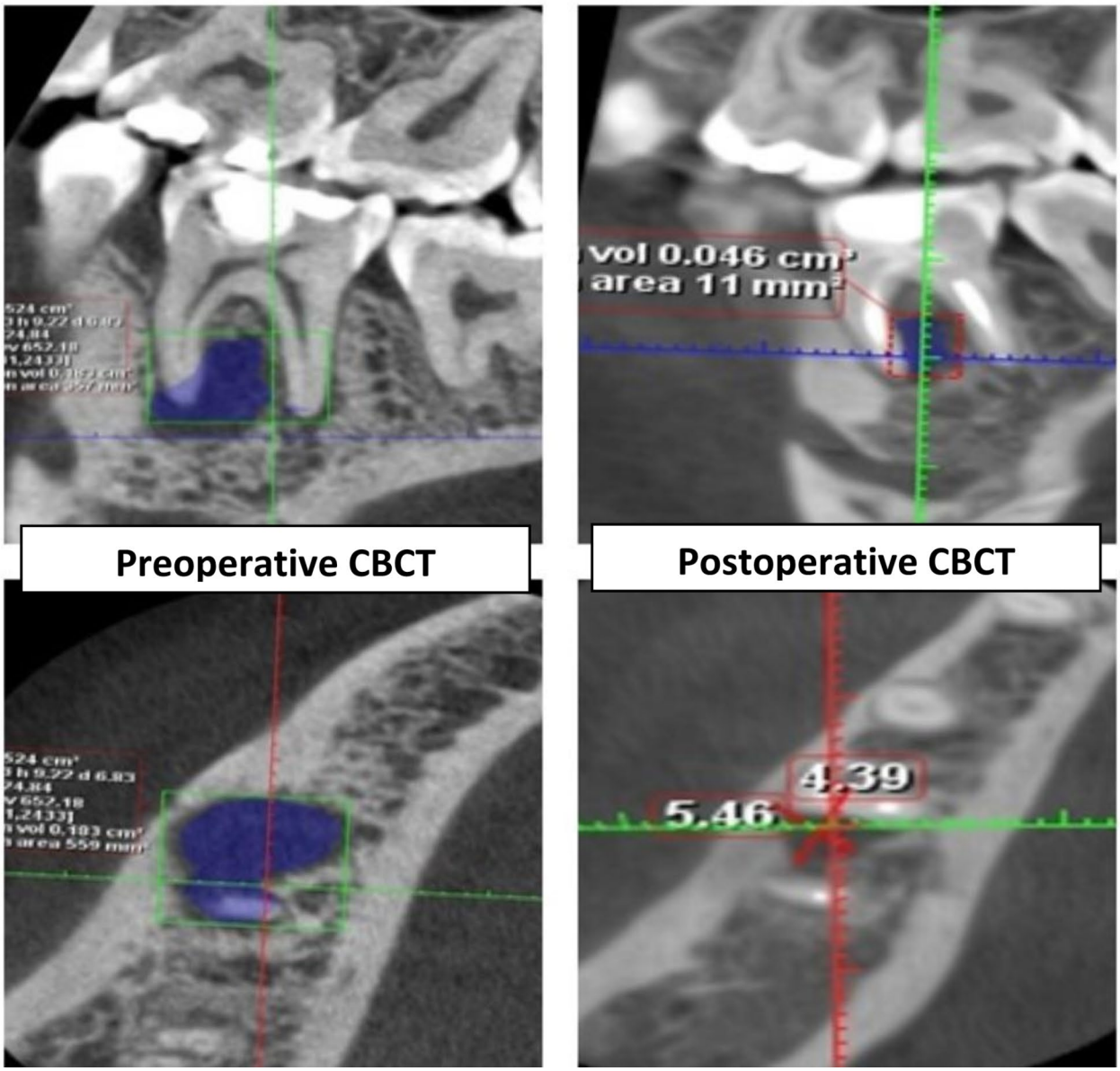


Fig. 3 CBCT images showing the difference in the volume of the periapical lesion in the ML group

and application approaches to identify the best possible protocol to use diode laser or PBM as an aid in healing. Our study contributes to the limited body of research exploring the impact of PBM on periapical bone healing in the context of endodontic treatment. Future research should aim to fill this gap by conducting more targeted studies on the specific effects of PBM on periapical bone healing following endodontic therapy. To enhance the understanding of PBM's impact on periapical healing, we recommend implementing shorter review intervals in study designs. Furthermore, utilizing more sophisticated methods, such as volumetric measurements or established CBCT periapical indices, will yield a more comprehensive and precise evaluation of healing outcomes. Lastly, we advocate for the exploration of AI-based software in conjunction with these traditional methods to improve lesion measurement and assessment.

Conclusion

The size of the periapical bony lesion decreased significantly after conventional endodontic treatment whether Diode Laser is applied or not.

Author contributions I.H.: study design, data acquisition, manuscript preparation, patient treatment, laser application, and follow-up. O.M.: subject enrolment, preparation of material and devices, RCT registra-

tion. H.E: data comparison, statistical analysis, and final revision.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).
There was no funding provided to the authors by any organization.

Data availability The data presented in this study are available upon request from the corresponding author.

Declarations

Ethical approval This study was designed and reported by adhering to the Consolidated Standards of Reporting Trials statement CONSORT. It was assessed and then approved by the Ethics Committee, Faculty of Dentistry, Ain Shams University (Approval code: FDASURecID091703). The study was performed per the Declaration of Helsinki.

Consent to participate Informed consent was obtained from all subjects and/or their legal guardian(s).Consent to publish The authors affirm that participants provided informed consent for the publication of the images in Figure(s) 2 and 3. All patients signed informed consent regarding publishing their data.

Consent to participate The authors have stated explicitly that there are no conflicts of interest in connection with this article.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Buonavoglia A, Zamparini F, Lanave G, Pellegrini F, Diakoudi G, Spinelli A, Lucente MS, Camero M, Vasinioti VI, Gandolfi MG, Martella V, Prati C (2023) Endodontic Microbial communities in Apical Periodontitis. J Endod 49(2):178–189. https://doi.org/10.1016/j.joen.2022.11.015
2. Orstavik D, Kerekes K, Eriksen H (1986) The periapical index: a scoring system for radiographic assessment of apical periodontitis. Endod Dent Traumatol 2(1):20–34. https://doi.org/10.1111/j.1600-9657.1986.tb00119.x
3. Prada I, Micó-Muñoz P, Giner-Lluesma T, Micó-Martínez P, Muwaquet-Rodríguez S, Albero-Monteagudo A (2019) Update of the therapeutic planning of irrigation and intracanal medication in root canal treatment. A literature review. J Clin Exp Dent 11(2):e185–e193. https://doi.org/10.4317/jced.55560
4. Chubb DWR (2019) A review of the prognostic value of irrigation on root canal treatment success. Aust Endod J 45(1):5–11. https://doi.org/10.1111/aej.12348
5. Gulabivala K, Ng YL (2023) Factors that affect the outcomes of root canal treatment and retreatment reframing of the principles. Int Endod J 56 Suppl 282–115. https://doi.org/10.1111/iej.13897
6. Metzger Z, Huber R, Slavescu D, Dragomirescu D, Tobis I, Better H (2009) Healing kinetics of periapical lesions enhanced by the apexum procedure: a clinical trial. J Endod 35(2):153–159. https://doi.org/10.1016/j.joen.2008.11.019
7. Kohli A, Anehosur V, Radder K (2023) Comparative study of the efficacy of low-level laser therapy and dexamethasone in reducing the severity of post-operative inflammatory response following Surgical extraction of Mandibular Third molars. J Maxillofac Oral Surg 22(1):165–171. https://doi.org/10.1007/s12663-021-01681-3
8. Pandeshwar P, Roa MD, Das R, Shastry SP, Kaul R, Srinivasreddy MB (2016) Photobiomodulation in oral medicine: a review. J Investig Clin Dent 7(2):114–126. https://doi.org/10.1111/jicd.12148
9. Guerreiro MYR, Monteiro LPB, de Castro RF, Magno MB, Maia LC, da Silva Brandão JM (2021) Effect of low-level laser therapy on postoperative endodontic pain: an updated systematic review. Complement Ther Med 57:102638. https://doi.org/10.1016/j.ctim.2020.102638
10. Kreisler MB, Haj HA, Noroozi N, Willershausen B (2004) Efficacy of low level laser therapy in reducing postoperative pain after endodontic surgery– a randomized double blind clinical study. Int J Oral Maxillofac Surg 33(1):38–41. https://doi.org/10.1054/ijom.2002.0449
11. Lopes LPB, Herkrath FJ, Vianna ECB, Gualberto Júnior EC, Marques AAF, Sponchiado Júnior EC (2019) Effect of photobiomodulation therapy on postoperative pain after endodontic treatment: a randomized, controlled, clinical study. Clin Oral Investig 23(1):285–292. https://doi.org/10.1007/s00784-018-2435-9
12. Lins RD, Dantas EM, Lucena KC, Catão MH, Granville-Garcia AF, Carvalho Neto LG (2010) Biostimulation effects of low-power laser in the repair process. Bras Dermatol 85(6):849–855. https://doi.org/10.1590/s0365-05962010000600011
13. Rocha Júnior AM, Vieira BJ, de Andrade LCF, Aarestrup FM (2007) Effects of low-level laser therapy on the progress of wound healing in humans: the contribution of in vitro and in vivo experimental studies. J Vasc Bras 6:257–265. https://doi.org/10.1590/S1677-54492007000300009
14. Rola P, Szymon W, Maciej L, Adrian D, Adrian W (2022) Changes in Cell Biology under the influence of low-level. Laser Therapy Photonics 9(7):502. https://doi.org/10.3390/photonics9070502
15. Minatel DG, Frade MA, França SC, Enwemeka CS (2009) Phototherapy promotes healing of chronic diabetic leg ulcers that failed to respond to other therapies. Lasers Surg Med 41(6):433–441. https://doi.org/10.1002/lsm.20789
16. Renno AC, McDonnell PA, Crovace MC, Zanotto ED, Laakso L (2010) Effect of 830 nm laser phototherapy on osteoblasts grown in vitro on Biosilicate scaffolds. Photomed Laser Surg 28(1):131–133. https://doi.org/10.1089/pho.2009.2487
17. Mozzati M, Martinasso G, Cocero N, Pol R, Maggiora M, Muzio G, Canuto RA (2011) Influence of super pulsed laser therapy on healing processes following tooth extraction. Photomed Laser Surg 29(8):565–571
18. Schulz KF, Altman DG, Moher D (2010) CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. J Pharmacol Pharmacother 1(2):100–107. https://doi.org/10.4103/0976-500x.72352
19. Abdelgawad LM, Kenawi LM (2016) Clinical and radiographic evaluation of healing effect of low level laser therapy on chronic apical lesions. Egyptian Dental Journal 1;62(1-January (Part 5)):1039-49. https://doi.org/10.21608/edj.2016.95418
20. Arslan H, Doğanay E, Karataş E, Ünlü MA, Ahmed HMA (2017) Effect of low-level laser therapy on Postoperative Pain after Root

Canal Retreatment: a preliminary Placebo-controlled, Triple-blind, randomized clinical trial. J Endod 43(11):1765–1769. https://doi.org/10.1016/j.joen.2017.06.028
21. Eroglu CN, Keskin Tunc S (2016) Effectiveness of single Session of low-level laser therapy with a 940 nm Wavelength Diode Laser on Pain, Swelling, and Trismus after impacted third molar surgery. Photomed Laser Surg 34(9):406–410. https://doi.org/10.1089/pho.2016.4101
22. Ismail HH, Obeid M, Hassanien E (2023) Efficiency of diode laser in control of post-endodontic pain: a randomized controlled trial. Clin Oral Investig 27(6):2797–2804. https://doi.org/10.1007/s00784-023-04864-z
23. Kim HS, Lee S, Kim JH (2018) Real-world evidence versus Randomized Controlled Trial: Clinical Research based on Electronic Medical records. J Korean Med Sci 26(34):e213. https://doi.org/10.3346/jkms.2018.33.e213
24. Slot DE, Jorritsma KH, Cobb CM, Van der Weijden FA (2014) The effect of the thermal diode laser (wavelength 808–980 nm) in non-surgical periodontal therapy: a systematic review and meta-analysis. J Clin Periodontol 41(7):681–692. https://doi.org/10.1111/jcpe.12233
25. Usumez A, Cengiz B, Oztuzcu S et al (2014) Effects of laser irradiation at different wavelengths (660, 810, 980, and 1,064 nm) on mucositis in an animal model of wound healing. Lasers Med Sci 29:1807–1813. https://doi.org/10.1007/s10103-013-1336-z
26. Fazlyab M, Esmaeili Shahmirzadi S, Esnaashari E, Azizi A, Moshari AA (2021) Effect of low-level laser therapy on postoperative pain after single-visit root canal retreatment of mandibular molars: a randomized controlled clinical trial. Int Endod J 54(11):2006–2015. https://doi.org/10.1111/iej.13608
27. Vase L, Wartolowska K (2019) Pain, placebo, and test of treatment efficacy: a narrative review. Br J Anaesth 123(2):e254–e262. https://doi.org/10.1016/j.bja.2019.01.040
28. Antony DP, Thomas T, Nivedhitha MS (2020) Two-dimensional Periapical, panoramic radiography Versus three-dimensional cone-beam computed Tomography in the detection of Periapical Lesion after Endodontic Treatment: a systematic review. Cureus 12(4):e7736. https://doi.org/10.7759/cureus.7736
29. Ahlowalia MS, Patel S, Anwar HM, Cama G, Austin RS, Wilson R, Mannocci F (2013) Accuracy of CBCT for volumetric measurement of simulated periapical lesions. Int Endod J 46(6):538–546. https://doi.org/10.1111/iej.12023
30. Karamifar K, Tondari A, Saghiri MA (2020) Endodontic Periapical Lesion: An Overview on the Etiology, Diagnosis and Current Treatment Modalities. Eur Endod J 14;5(2):54–67. https://doi.org/10.14744/eej.2020.42714
31. Mittal P, Logani A, Shah N, Pandey RM (2016) Effect of apical clearing technique on the treatment outcome of teeth with asymptomatic apical periodontitis: a randomized clinical trial. J Conserv Dent 19(5):396–401. https://doi.org/10.4103/0972-0707.190006
32. Tsesis I, Faivishevsky V, Kfir A, Rosen E (2009) Outcome of surgical endodontic treatment performed by a modern technique: a meta-analysis of literature. J Endod 35(11):1505–1511. https://doi.org/10.1016/j.joen.2009.07.025
33. Tang CQY, Chiow SM, Lai SHS, Chia DSY (2022) The Effect of Hand Dominance, Age, gender, fracture comminution and ASA Status on Time to Fracture Healing following Surgical fixation of distal Radius fractures. J Hand Surg Asian Pac 27(3):459–465. https://doi.org/10.1142/s2424835522500461
34. Chaves ME, Araújo AR, Piancastelli AC, Pinotti M (2014) Effects of low-power light therapy on wound healing: LASER x LED. Bras Dermatol 89(4):616–623. https://doi.org/10.1590/abd1806-4841.20142519
35. Noba C, Mello-Moura ACV, Gimenez T, Tedesco TK, Moura-Netto (2018) Laser for bone healing after oral surgery: systematic review. Lasers Med Sci 33(3):667–674. https://doi.org/10.1007/s10103-017-2400-x
36. Kazem Shakouri S, Soleimanpour J, Salekzamani Y, Oskuie MR (2010) Effect of low-level laser therapy on the fracture healing process. Lasers Med Sci 25(1):73–77. https://doi.org/10.1007/s10103-009-0670-7
37. Posten W, Wrone DA, Dover JS, Arndt KA, Silapunt S, Alam M (2005) Low-level laser therapy for wound healing: mechanism and efficacy. Dermatol Surg 31(3):334–340. https://doi.org/10.1111/j.1524-4725.2005.31086
38. Kulkarni S, Meer M, George R (2019) Efficacy of photobiomodulation on accelerating bone healing after tooth extraction: a systematic review. Lasers Med Sci 34(4):685–692. https://doi.org/10.1007/s10103-018-2641-3
39. Topçuoğlu HS, Akpınar B (2021) The effect of low-level laser therapy on the success rate of inferior alveolar nerve blocks in mandibular molars with symptomatic irreversible pulpitis: a randomized clinical trial. Int Endod J 54(10):1720–1726. https://doi.org/10.1111/iej.13596
40. Asnaashari M, Safavi N (22013) Application of low-level lasers in Dentistry (Endodontic). J Lasers Med Sci 4(2):57–66. https://doi.org/10.22037/JLMS.V4I2.4187
41. Metin R, Tatli U, Evlice B (2018) Effects of low-level laser therapy on soft and hard tissue healing after endodontic surgery. Lasers Med Sci 33(8):1699–1706. https://doi.org/10.1007/s10103-018-2523-8
42. Naseri M, Asnaashari M, Moghaddas E, Vatankhah MR (2020) Effect of low-level laser therapy with different locations of Irradiation on Postoperative Endodontic Pain in patients with symptomatic irreversible pulpitis: a double-blind randomized controlled trial. J Lasers Med Sci 11(3):249–254. https://doi.org/10.34172/jlms.2020.42
43. Asnaashari M, Ashraf H, Daghayeghi AH, Mojahedi SM, Azari-Marhabi S (2017) Management of Post Endodontic Retreatment Pain with Low Level Laser Therapy. J Lasers Med Sci 8(3):128–131. https://doi.org/10.15171/jlms.2017.23
44. Turhani D, Scheriau M, Kapral D, Benesch T, Jonke E, Bantleon HP (2006) Pain relief by single low-level laser irradiation in orthodontic patients undergoing fixed appliance therapy. Am J Orthod Dentofac Orthop 130(3):371–377. https://doi.org/10.1016/j.ajodo.2005.04.036

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.